

The next phase of identifying illicit activity in Bitcoin

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Abstract

Identifying illicit behavior in the Bitcoin network is a well-explored topic. The methods proposed over time have generated great insights into the deanonymization of the Bitcoin user base through the clustering of inputs and outputs. With advanced techniques being deployed by Bitcoin users, these heuristics are now being challenged in their ability to aid in the detection of illicit activity. In this paper, we provide a comprehensive list of methods deployed by malicious actors on the network and illicit transaction mining methods. We detail the evolution of the heuristics that are used to deanonymize Bitcoin transactions. We highlight the issues associated with conducting law enforcement investigations and propose recommendations for the research community to address these issues. Our recommendations include the release of public data by exchanges to allow researchers and law enforcement to further protect the network from malicious users. We recommend the enhancement of current heuristics through machine learning methods and discuss how researchers can take the fight head-on against expert cybercriminals.

KEYWORDS

blockchain, cryptocurrency, cybersecurity, privacy

1 | INTRODUCTION

Financial cybercrime has become more prevalent in the crypto-asset space over the past 10 years.¹ There are highly publicized cryptocurrency frauds and hacks resulting in tens of millions worth of dollars being stolen and subsequently laundered through the various cryptocurrency networks using multiple blockchains and privacy-enhancing tools made available to them publicly. There have been high-profile cases of cybercriminals demanding cryptocurrency in return for the decryption of data assets in ransomware attacks^{2–4} and the laundering of FIAT money through cryptocurrency using techniques to obfuscate the origin of funds. In order to legitimize the proceeds of criminal activity, criminals must launder their illegally obtained funds. The increase in crime equally causes an increase in law enforcement involvement in the identification of illicit activity on the networks. The challenges we discuss lie in identifying the advanced malicious users of Bitcoin. Individuals and organizations who have in-depth, expert knowledge of cryptocurrency can take advantage of nuances unknown to the common user. In order to defend or identify weaknesses in tools provided to criminals, it is necessary to understand how they work.

Let us take the example of the recent case of The Wall Street Market (WSM),^{5,6} launched in 2016 and taken down in May 2019 as the result of a multi-agency effort. The WSM was maintained by three administrators, Klaus-Martin

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Frost, Jonathan Kalla, and Tibo Lousee, who law enforcement agencies managed to trace, identify, and capture. Identifying the third administrator, “Frost” was possible mainly by tracing the cryptocurrency transactions tied to wallets owned by the individual.⁶ The US Postal Inspection Service recognized the Bitcoin wallets used by Frost by analyzing blockchain transactions and the information obtained from the *proprietary software*. At a high level, the identification workflow included the following: discovering the wallets, tracing the payments between wallets, reversing the mixing process of the tainted transactions, and finally associating the wallets with darknet marketplaces. The cross-contamination between cryptocurrency and wallets played a crucial role in the discovery process. It is unclear how USPS discovered the cross-contamination; we can assume that it employs current state-of-the-art techniques, such as address clustering based on multi-input heuristics. However, recent services such as Samurai⁷ can bypass the detection and may result in false positives. There is a gap in understanding detection methods and their robustness with changes in the crypto landscape relevant to investigations. In this systematization of knowledge, we aim to fill this gap by documenting a comprehensive list of illicit transaction mining methods and highlighting their issues when conducting legal/forensics investigations. We also propose recommendations for the research community to address the identified problems.

Our contributions to the research community are as follows:

- Provide comprehensive and easily digestible methods of Bitcoin obfuscation.
- Detail the evolution and research of Bitcoin heuristics used to deanonymize the network.
- Highlight the issues and drawbacks of the current heuristic methods deployed in Bitcoin to deanonymize users.
- Discuss potential improvements to the heuristics through machine learning and deep learning techniques.
- Explain the privacy enhancing tools made available to Bitcoin users aiding in the evasion of currently deployed heuristics.
- Recommendations including the release of attribution data to aid in the identification of criminal activity and aid in the protection of the Bitcoin user base, combining artificial intelligence methods to enhance heuristics, and how to deal with the sophisticated cybercriminal.

In Section 2, we provide an overview of cryptocurrency, the unspent transaction output (UTxO), and the money laundering techniques deployed by criminals in Bitcoin. In Section 3, we discuss the identification methods currently deployed in Bitcoin. This includes a thorough analysis of the Bitcoin heuristic research. We also discuss about the other data sources that can improve identification and researched identification methods being published. We also discuss requirements for proposed methods to be introduced into law enforcement. In Section 4, we cover several of the privacy enhancing tools that are being used by cryptocurrency users to obfuscate and evade detection on the network by the identification methods covered in the paper. Finally, Section 5 covers our recommendations to the research community in tackling the next phase of illicit activity identification in Bitcoin.

2 | OVERVIEW

2.1 | Related work

Other SoKs and surveys have tackled the area of illicit activity in Bitcoin, but to the best of our knowledge, none have addressed the enhancement of heuristics through artificial intelligence nor included detailed recommendations surrounding anomaly detection application in the field of identifying illicit activity in Bitcoin. There has been several articles that have covered the domain of Bitcoin deanonymization, cryptocurrency money laundering, privacy, and Bitcoin heuristics.^{8–12}

2.2 | Cryptocurrency

Cryptocurrency is a digital asset that represents value, typically denominated in dollars or euros. It has the ability to be transferred in cyberspace via blockchain technology that uses computational techniques to ensure the authenticity of the transactions. It bypasses traditional financial institutions that would act as the intermediary in settlements between two transacting parties. Bitcoin is the most valued cryptocurrency currently being used and the most globally known.

The history of cryptocurrency and the underlying mechanics of Bitcoin are well documented in several surveys and SoK papers.^{13–15}

Bitcoin transactions have offered unique methods of transferring capital and allow users to collude in pools of transactions to further obfuscate any deterministic links between input and output for any given transaction, thus enhancing the user's privacy.

2.2.1 | Unspent transaction output (UTxO)

The uniqueness of Bitcoin transactions against traditional FIAT-based transactions is the use of the Unspent Transaction Output (UTxO). The UTxO represents output yet to be spent by Bitcoin users who have either been transferred to them through a previous transaction from another address or were rewarded them in newly minted blocks via a coin-base transaction.

Cryptocurrencies operating through UTxO methods include Bitcoin, Litecoin, Zcash, and Monero. UTxO offers additional privacy to users by offering Bitcoin users the ability to send the change received in a transaction to a new address, to disassociate the original sender from the change output. Having different change addresses available makes it difficult for third-party observers to differentiate an output as being either the change of a transaction that is being returned to the original sender or whether the sender has split a single input into multiple outputs. For example, Alice, *A*, wishes to send 0.75BTC of 1.00BTC to another user, Bob, *B*. The change is returned to Alice as a new unspent transaction, *A_b*, which would be 0.25BTC. This would appear to a spectator as a single input and two outputs. This is demonstrated in Figure 1 where two scenarios are shown. The first scenario demonstrates the change of output being returned to a new address, *A_b*. The second shows one input being split into two outputs being sent to two of Bob's different addresses, *B₁* and *B₂*. Without attribution information of the underlying wallets, it can be very difficult to verify the destination of outputs. Many heuristics applied to the blockchain to deanonymize or identify users use clustering methodology on the UTxO set.

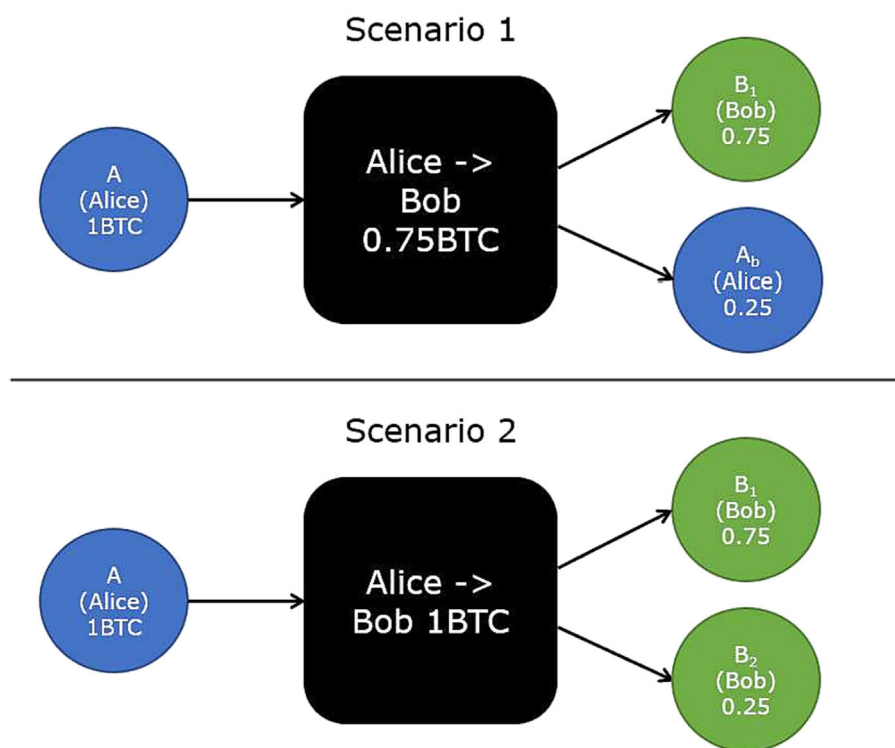


FIGURE 1 An example of the UTxO change operation returning 0.25BTC of change *A_b* to Alice, *A*, and transferring 0.75BTC to Bob, *B*. The second scenario shows a more sophisticated Alice splitting a single input into two outputs, both to Bob, *B₁* and *B₂*

2.3 | Cryptocurrency money laundering

To effectively combat malicious activity on the Bitcoin network, it is necessary to understand the methods and tools available to criminals that mask their activities. Criminals launder their illegally obtained funds by layering transactions in the Bitcoin network and hiding their identities through transaction mechanics like the UTxO change addresses explained in Section 2.2.1.

Money laundering is comprised of three steps: (i) placement, (ii) layering, and (iii) integration. Placement is the process of depositing criminals proceeds into a financial system. Layering consists of moving money through complicated tranches of transactions with the goal to obfuscate the origin. Finally, integration is when the criminal proceeds have been reabsorbed into the public economy through purchases of luxury goods, property, or investments.¹ An enticing factor of money laundering with the use of cryptocurrency is the associated characteristics with digital assets. These include decentralization and perceived transaction anonymity.¹⁶ Highly descriptive papers have been published examining the strengths and weaknesses of the obfuscation techniques deployed on Bitcoin.^{9,10,17}

2.3.1 | Mixing

Mixing is the process of obfuscating the origin of targeted inputs by blending them with many other Bitcoin from coluding parties. The user can access a mixing service website and is requested to send their Bitcoin to a targeted address, along with a prescribed mixing fee. The user receives their newly mixed Bitcoin in a new wallet with obfuscated connections to the originating account. The received Bitcoin (BTC) is not necessarily the original BTC that the user has sent to the network. ChipMixer used a clever technique of providing users with *chips* that were redeemable on the platform. A user may receive 10 chips for 1BTC for example, and the user can redeem those chips at some point in the future in multiple or a single order. These chips are converted into BTC by the platform and the user will receive BTC that may have been sent to the service at a much earlier point in time. This means an investigator attempting to trace a transaction that enters into ChipMixer will have difficulty in following the funds through the blockchain as the chips could be converted to BTC and the user will have spendable inputs from much earlier transactions in the blockchain. This example is demonstrated in Figure 2.

Tracing transactions that interact with a mixer can be difficult to track due to the layered process in which the services act. There can be transactions with hundreds of inputs and hundreds of outputs making the identification of

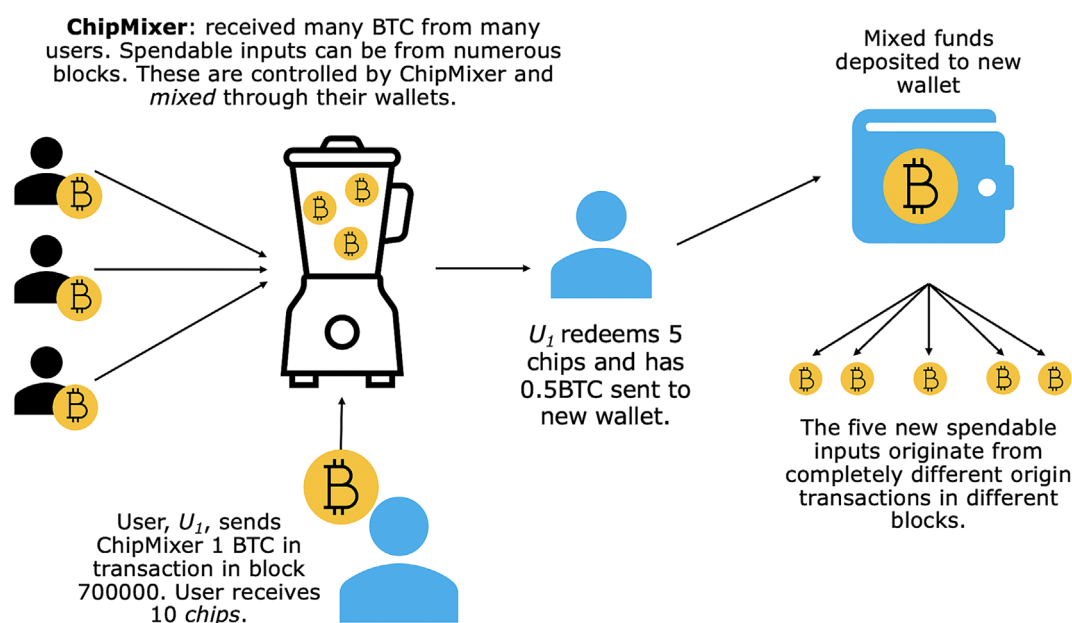


FIGURE 2 Detailed example of a ChipMixer transaction.

origin and destination a difficult task. There are three forms of mixing described by Crawford et al.¹⁸ and Wu et al.¹⁹ These include centralized, decentralized, and cross-blockchain mixing. Centralized requires the use of a trusted third party that acts as the custodian of the funds to be mixed. These are the dominant form making up approximately 90% of the mixing market.¹⁸ An example of a user journey using a centralized mixing service is shown in Figure 3.

Decentralized platforms are algorithmically driven and select random colluding users to reach a consensus on mixing transactions. The cross-blockchain mixing service allows the exchange of one cryptocurrency, like Bitcoin, with other cryptocurrencies such as Ether¹⁹ through the use of smart contracts. This increases the difficulty in tracking origin and destination points. An example of each mixing process is shown in Figure 4.

2.3.2 | Coinjoin

Coinjoin transactions are collaborative multisignature transactions where multiple users combine inputs in a single transaction to specified target outputs. Identifying a coinjoin transaction against a regular one can be difficult without underlying attribution information on wallets.

An example of a coinjoin, shown in Figure 5, can be explained by comparing two transactions. In two separate transactions there are three inputs. In the first transaction, three inputs, $Input_A$, $Input_B$, and $Input_C$ belong to the same person. They are sent in a single output to $Output_A$. This is considered a normal transaction (Tx) in this example. The second transaction is a coinjoin, where three users have collaborated together using a multisignature process to each send an individual input into a transaction, and send them to three different output addresses. This transaction is very difficult to deanonymize, and this transaction breaks heuristic assumptions. Further confusing a coinjoin is interjecting new change addresses to further obfuscate the transaction.

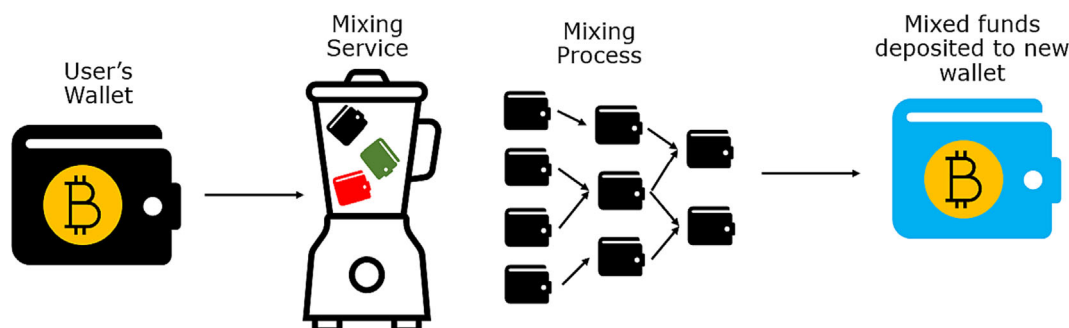


FIGURE 3 High level example of centralized mixing service.

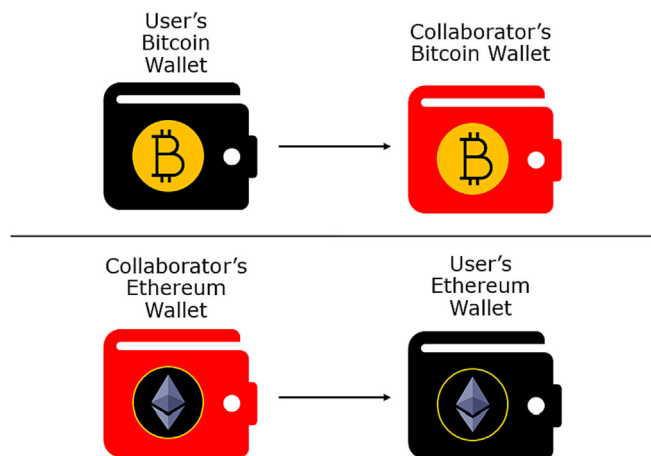


FIGURE 4 An example of cross-blockchain mixing demonstrating two users who wish to exchange different cryptocurrencies.

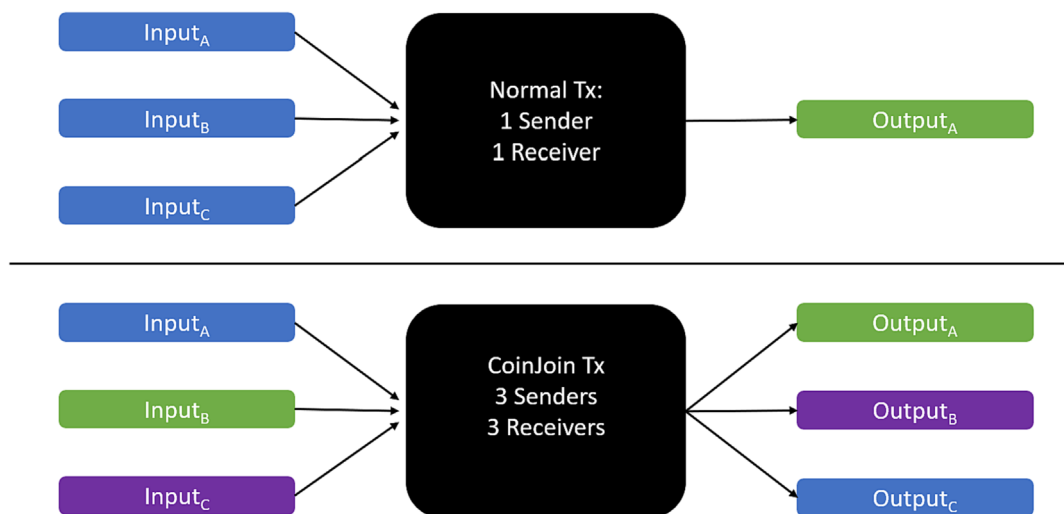


FIGURE 5 The figure compares a normal transaction to a coinjoin transaction. The different colors represent each user in the transaction.

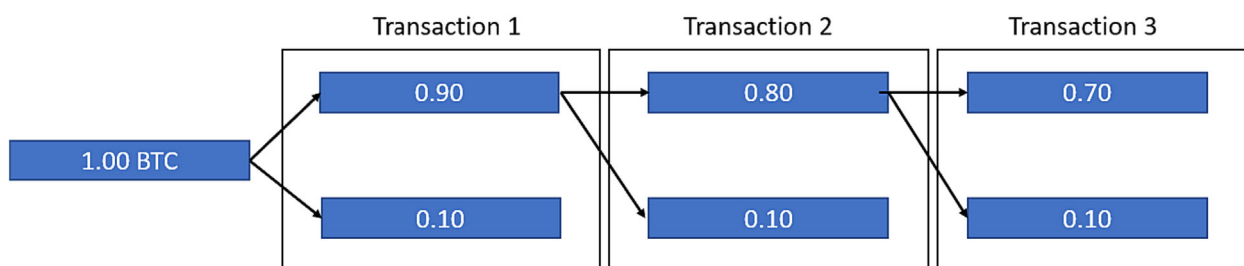


FIGURE 6 An example of a peeling chain generating six new addresses through three transactions from a single input.

2.3.3 | Peeling

Peeling is a method of iteratively sending a small amount of Bitcoin to different addresses and sending the change to new addresses all owned by the same entity or person. This is a method of “peeling” away small chunks of Bitcoin to many wallets and creating a vast layering process that is hard to aggregate and identify the underlying owner. An example of a peeling chain begins with a single input of 1BTC. This input is split into two outputs of 0.90 and 0.10 BTC in the first transaction and iteratively layered taking small amounts of BTC through the next transaction, taking advantage of newly generated change addresses. In this example, 0.10BTC is peeled from each transaction to produce six new addresses on the blockchain from a single input address. This is shown in Figure 6.

3 | IDENTIFICATION METHODS

In this section, we discuss the current identification methods deployed to deanonymize the Bitcoin blockchain. We explain and discuss the various heuristics that are applied to the Bitcoin blockchain. We provide a comprehensive review of the research in heuristics. We explain the ease of use for applying heuristics to the blockchain and how reliable they are when deployed.

3.1 | Heuristics

Heuristics are defined processes created to deanonymize Bitcoin users. Heuristics have shown to be a dominant force in research and industry in identifying illicit activity through the clustering of inputs and outputs in an attempt to identify

the controlling entity or individual. However, with advancements in the tools and methods deployed by advanced cryptocurrency users, heuristics are beginning to wane in their efficacy. Knowledge of the underpinning mechanics of heuristics is quite public. With the same thought of defenders needing to know attackers techniques, attackers take great advantage of finding weaknesses in the defensive mechanisms deployed by exchanges and law enforcement agencies. Bitcoin Heuristics have been researched and published as early as 2013 by Reid et al¹⁵ and Meiklejohn et al²⁰ independently, which both acted as a cornerstone for future heuristic development and being the basis for products offered by numerous cryptocurrency intelligence companies such as Chainalysis (<https://www.chainalysis.com>), Elliptic (<https://www.elliptic.co>), and CipherTrace (<https://ciphertrace.com>).

Heuristic methodologies are highly documented in literature.^{15,21-23} Heuristics are relatively easy to deploy with libraries available such as BlockSci.²⁴ The user needs to have basic programming knowledge of Python and the resources required to download their own Bitcoin node through the Bitcoin core at <https://bitcoincore.org>. This process has been made simple with the use of a GUI application and clear instructions. BlockSci can be applied to the locally stored Bitcoin blockchain or RPC to a hosted instance. BlockSci allows for fast parsing of the blocks, stored as dat files, which requires little to no manual processing of information by the users. BlockSci's main feature is the ability to perform "clustering" or aggregation of addresses through heuristics. BlockSci gives a low barrier to entry for users who want to track flows of Bitcoin through the blockchain with the ability to deanonymize particular users with additional labeled data that could be extracted easily from <https://www.walletexplorer.com>, who perform their own clustering techniques to track users with the injection of label data that they make publicly available.

3.2 | Multi-input heuristic

Multi-input scenarios are created based on the underlying mechanics of the UTxO set. A Bitcoin user can have access to many inputs. If a user transacts with another user, they send multiple inputs to match the desired output; if the inputs exceed the desired output, the remainder is sent as an output via change. For example, if Alice has three inputs valued at 0.33BTC each and wants to send Bob 1BTC, then Alice will send three 0.33BTC inputs. This type of transaction is shown in Figure 7.

The multi-input heuristic tries to cluster the inputs in a transaction and link them to a controlling entity or individual. In our example from above, the three inputs would be assigned to the same controlling user, Alice. Figure 7 demonstrates how the multi-input heuristic attempts to classify inputs as being from the same sender.

There are multiple variations of the multi-input heuristics that have evolved over time to address protocol changes, attempt adjustments to the process to more correctly identify clusters, and prevent scenarios of super clusters being created with labeling of exchange addresses and services.

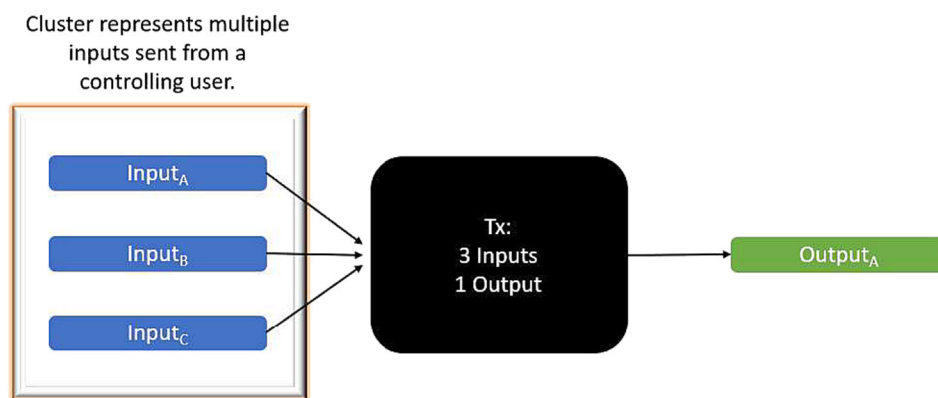


FIGURE 7 In this example, the multi-input heuristic will cluster all of the inputs as belonging to the same controlling user. This heuristic is broken when the transaction is performed through a multi-signature process such as a Coinjoin.

3.3 | Change address detection

As described in Section 2.2.1, change is the output sent back to the original sender. Change address detection tries to classify one of the outputs as a change in a standard transaction. In Figure 1, the heuristic would correctly identify the 0.25BTC as Alice's change in the first scenario. This would allow us to build up a profile of Alice's balance and mark that change as a future input in a transaction to track the spending.

The underlying assumption in the change address detection is that there is one user sending Bitcoin, and one user receiving Bitcoin. Or sufficient attribution of addresses is known for multiple outputs leaving a single output to be determined as change.

To expand multi-input heuristic to capture the entire blockchain and enhance tracking and deanonymization, the change address heuristic attempts to identify an output belonging to the original sender. The multi-input combined with the change address heuristic then captures both sides of the transaction, and the newly labeled change address can be used in the multi-input heuristic on the next transaction performed by the tracked user. An example of this would be the following: A transaction, Tx 1, has three inputs ($Input_A, Input_B, Input_C$) and two outputs ($Output_A, Output_B$). Two heuristics are combined to track the user. Multi-input heuristic performs clustering/aggregation of the inputs and assign that cluster an identifier. The change heuristic then labels one of the outputs as change. This change output will be assigned the same identifier as the original multi-input cluster. In the following transaction, Tx 2, which occurs at $Time_{Tx1} + Time_{block+1}$, $Output_B$ is used in a new transaction with two different input addresses, $Input_D$ and $Input_E$. As $Output_B$ has been assigned a cluster identifier, the multi-input heuristic will then cluster the inputs of Tx 2, and due to $Output_B$ being included in the inputs, that cluster will assign the same identifier from Tx 1 inputs and $Output_B$. In Tx 2, $Output_D$ has been assigned as the change address and will be labeled with the same cluster identifier.

This process is demonstrated in Figure 8.

To highlight and contrast the various heuristics published in literature, we present a thorough list of existing techniques with their nuanced differences below:

1. **Paper:** Quantitative Analysis of the Full Bitcoin Transaction Graph²⁵

- (a) **Tool/Algorithm Outline:** The quantitative analysis performed by Ron et al. is a small, early snapshot of the Bitcoin blockchain as the paper was published in 2012. The techniques were novel in their method of tracking user behavior and analyzing Bitcoin acquisition and spending by the user base. More importantly, they

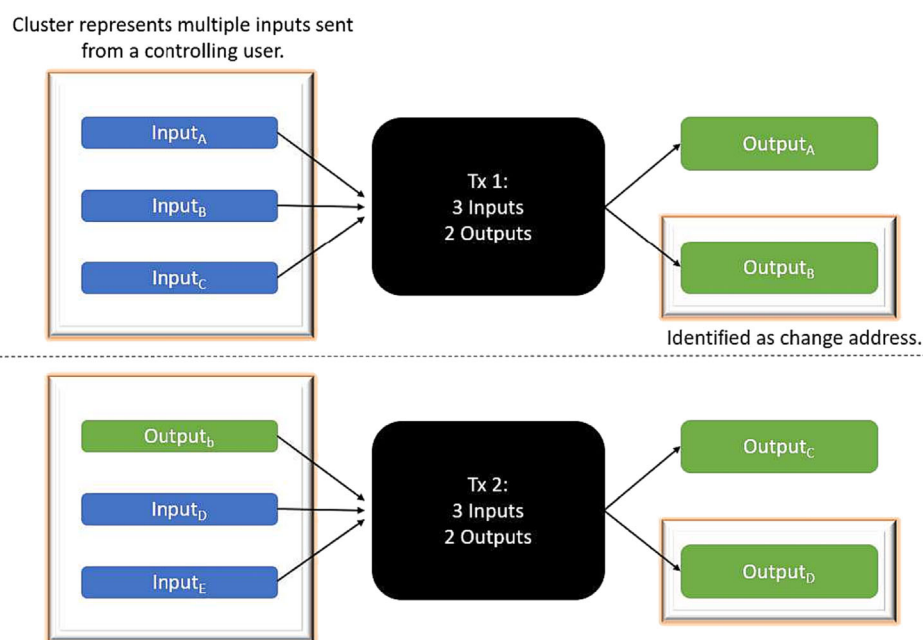


FIGURE 8 The multi-input heuristic combined with the change address heuristic can improve deanonymization of the Bitcoin blockchain.

analyzed the techniques users deployed to move Bitcoin between accounts to improve their privacy on the network.

- (b) **Heuristic Description:** The methodology performed in this quantitative analysis was not strictly labeled as heuristic but aggregates addresses (inputs and outputs) in a very similar manner. The authors reduced the transaction graph of the first 180,000 blocks through a variant of the Union-Find graph algorithm.²⁶ They call the full, granular Bitcoin transaction graph the *address graph* and the aggregated, contracted transaction graph the *entity graph*. In their reduced entity graph, each transaction is considered to have a single sender and at most n receivers. This method can be seen as an early variant of the multi-input heuristic, aggregating the sender addresses into a single cluster or node on the transaction graph. The change identification is dealt through multi-input analysis, and they provide an example of a transaction in which address 1 and address 2 are used as inputs in transaction 1. Then, another transaction in which addresses 2 and 3 are used as inputs in transaction 2, it can be concluded that all three addresses are jointly owned by the entity.
2. **Paper:** An Analysis of Anonymity in the Bitcoin System¹⁵
 - (a) **Tool/Algorithm Outline:** Reid and Harrigan's¹⁵ early analysis of the Bitcoin blockchain challenged the anonymity of Bitcoin. In this paper, they introduce the concept of multi-input heuristic at roughly the same time as Meiklejohn et al.²⁰ Through topological structuring analysis of the Bitcoin transaction graph, Reid and Harrigan demonstrate the one-to-many mapping between the Bitcoin users and the public-keys, and they include additional information through fingerprinting to aid in the deanonymization of Bitcoin users.
 - (b) **Heuristic Description:** Reid and Harrigan model the Bitcoin transactions as a graph and labeled the user network. Essentially, this model is an early representation of the multi-input heuristic, where the inputs are clustered into a single vertex or node, and then a weighted directed edge is used to represent a single user spending an amount of Bitcoin in a transaction through multiple inputs. Coupled with the implementation of multi-input heuristic, they perform anonymity analysis on the privacy of the Bitcoin network by integrating off-network information through data collection of fingerprinting data available on Bitcoin transactions such as IP addresses and graph analysis techniques on their created transaction user network.
3. **Paper:** Evaluating User Privacy in Bitcoin²⁷
 - (a) **Tool/Algorithm Outline:** Similar to Reid and Harrigan's¹⁵ study on Bitcoin anonymity, Androulaki et al.²⁷ evaluate the privacy provided by Bitcoin in an experimental environment, which is a university setting. They introduced the concept of the shadow address heuristic. A shadow address is defined as the address created in a Bitcoin transaction that returns the change to a user when excess is paid to the target user
 - (b) **Heuristic Description:** Androulaki et al. implement the multi-input heuristic with reference to Reid and Harrigan's paper.¹⁵ The second heuristic implemented is their contribution, which is the shadow address identification. They use a simple method of labeling the unknown output address not seen before on the blockchain as a change or shadow address belonging to the original sender in the transaction. In this early study, they state that users rarely send Bitcoin to two different parties in a single transaction.
4. **Paper:** A Fistful of Bitcoins: Characterizing payments among men with no names²⁰
 - (a) **Tool/Algorithm Outline:** In this work, Meiklejohn et al.²⁰ introduce the concept of the change address heuristic. The creation of a transaction network is performed in a similar manner to Reid and Harrigan¹⁵ where they use a multi-input heuristic to create a single cluster of a sole entity owning the various inputs of a transaction. However, the key contribution is the change address heuristic that aims to classify one of the outputs as belonging to the original sender.
 - (b) **Heuristic Description:** The change address heuristic is an extension of the shadow address heuristic explored by Androulaki et al.²⁷ The extension focused on the emergence of users sending Bitcoin in a single transaction to more than one person, which bypasses many of the assumptions held in earlier heuristics attempting to classify the destination user. The reasoning behind this was to prevent the super cluster formation if numerous change addresses are erroneously labeled by the algorithms. This would cause future classification of users to be incorrect and snowball into a super cluster of Bitcoin address ownership. The objective of this heuristic was to radically reduce the false positive labels of change addresses. A key objective of identifying a change address in the Bitcoin network is to confirm that it is the first time it has been seen by the network to receive Bitcoin, apart from coinbase transactions.
5. **Paper:** Bitiodine: Extracting intelligence from the Bitcoin network²⁸
 - (a) **Tool/Algorithm Outline:** BitIodine is a collection of modules that allows the automatic parsing of the Bitcoin blockchain, clustering of addresses through two heuristic processes, classifies the addresses and users

through an indexing process, creates graphs through Gephi, and allows the export and visualization of the compiled information.

- (b) **Heuristic Description:** The heuristics are applied on the parsed blockchain data through a “clusterizer” module that attempts to aggregate the groups of addresses belonging to the same users. This combines multi-input and change address heuristics. The **multi-input heuristic** is a simple implementation that takes a set of inputs. If the cardinality of the set, S_i , is >1 , then all input addresses of that transaction belong to the same sender in the transaction. The inputs are assigned to a known or unknown user and are indexed in a larger user set u_k .

The second heuristic applied to the transaction where possible is the **shadow address guessing**, which handles the identification of change. There are two sections to the heuristic: The first is a deterministic stage, which states that if there are two outputs, and one of the two has never been seen on the Bitcoin blockchain before, then that is labeled as the change address and is indexed in the user set, u_k . The second section of the heuristic abuses an early bug in the Bitcoin client, which automatically assigned the change address as the first output of a transaction if there was greater than one output. This bug has since been amended after transactions occurring from February 3, 2013. Therefore, BitIodine implements the first section of the shadow address heuristic on identifying change addresses in transactions after February 3, 2013.

The shadow address guessing heuristic allows for the future indexing of multi-input heuristics when an input contains change addresses previously labeled in u_k to expand on a user set of identified addresses.

6. **Paper:** Structure and Anonymity of the Bitcoin Transaction Graph²⁹

- (a) **Tool/Algorithm Outline:** This paper by Ober et al. analyzes the methods deployed to increase anonymity in the Bitcoin network in 2013. The first 215,399 blocks were retrieved and analyzed to study global properties of the transaction graph. They focused on deanonymizing users who try to obfuscate themselves through a global passive adversary that attempts to link transactions or infer user identities.
- (b) **Heuristic Description:** Ober et al. also use the assumption presented by Reid and Harrigan,¹⁵ where the addresses used at input are considered to be controlled by the same entity. The idea of linkability of transactions is presented in this work through an event coined as a “merging event.” A merging event is when the transactions can be merged to be owned by an entity through linked outputs and inputs identified through the heuristic.

7. **Paper:** Could Network Information Facilitate Address Clustering in Bitcoin?³⁰

- (a) **Tool/Algorithm Outline:** This article by Neudecker and Hartenstein assesses the ability of deanonymizing the network by combining blockchain and network information to facilitate the Bitcoin address clustering process. The ambition of the research is to use the existing heuristics with IP address information that the authors extracted from the message flooding process of the Bitcoin network.
- (b) **Heuristic Description:** The methodology of this procedure is to analyze the address clusters that have been created using established heuristics discussed in this paper that are correlated to IP addresses associated to transactions based on Bitcoin network observations. The multi-input heuristic is deployed, and the change address identification is used by deploying the heuristic presented by Meiklejohn et al.²⁰; also implemented in the change address heuristic is the methodology published by Nick³¹ that covers *value based: Optimal Change* and *consumer based: redeeming transaction*. They also implement a method of controlling *cluster growth* presented by Harrigan and Fretter.³² The *value based: Optimal Change* works under the condition where a transaction has a single output but the value is smaller than any of its inputs, then this output is designated as change. The *consumer based: redeeming transaction* is a heuristic used for clustering consumer wallets that displayed a characteristic of re-using a change output as an input in a transaction. Finally, the *cluster growth* mechanism is a way to guard the clusters from merging into a super cluster. If two large clusters merge due to a new transaction, then it is likely a false positive and cluster growth mechanism tries to stop that.

The IP addresses are associated with transactions through heuristic methods created by the authors, Neudecker and Hartenstein.³⁰ The heuristic process they deploy involves the use of two monitor peers. A monitor peer “maintains a connection to all reachable peers in the network and log for each transaction.” In a Bitcoin transaction, there is an originator that sends a transaction to the researcher’s monitor peer. There are three rules of the heuristic to link an IP address to a transaction through the use of the information received by the monitor peer. They are detailed in the paper, and all three are associated with analyzing the time between transactions interacting with the monitor peers to decide if an IP address will be linked with a transaction or not.

8. **Paper:** Characterizing Entities in the Bitcoin Blockchain³³
 - (a) **Tool/Algorithm Outline:** The study performed by Jourdan et al.³³ examined the entities in Bitcoin transactions through analysis of the information revealed in entity transaction neighborhoods. They demonstrate that particular characteristics extracted through data mining and not controllable by the Bitcoin users and leaves open identifiable properties of entities on the network. This paper is one of the earlier studies into the application of machine learning models for the classification of Bitcoin entities.
 - (b) **Heuristic Description:** To create an entity graph, they implemented the common spending heuristic or multi-input. The entity graph can be seen as a similar technique to Reid and Harrigan's¹⁵ *user network*. The study also includes exchange entity classification through motifs introduced by Ranshous et al.³⁴ Jourdan et al. coupled their feature extraction with machine learning modeling (logistic regression and Light GBM) to classify address entities such as exchange, gambling, mining, services, and darknet marketplaces.
9. **Paper:** Automatic Bitcoin address clustering²³
 - (a) **Tool/Algorithm Outline:** In this paper, Ermilov et al.²³ look to expand on the multi-input and change address heuristics by introducing off-chain information in the address separation procedure for the address clustering stage. Their ambition is to enhance existing heuristics to address the drawbacks that allow for misclassification or evasion and improve deanonymization on the Bitcoin network.
 - (b) **Heuristic Description:** Ermilov et al. implement the multi-input heuristic and change address heuristic. Their off-chain information included in the clustering process is tag collection and negative pairs. Tag collection is simply scraping or manually identifying addresses and the controlling entity. The theory behind negative pairing is that it is unlikely that a transaction including an address directly from a darknet marketplace would be sent or received by an exchange. Therefore, the heuristic enhancement sees a logical rule and voting procedure integrated to the heuristics to improve the clustering process.
10. **Paper:** Heuristic-Based Address Clustering in Bitcoin²¹
 - (a) **Tool/Algorithm Outline:** This study by Zhang et al. re-examines the change address heuristic by introducing an enhancement detection method for identifying one-time change addresses. This heuristic is to aid the multi-input and change address heuristic where they cannot identify one-time change addresses. They label their heuristic as *Address Reuse-Based Change Address Detection*.
 - (b) **Heuristic Description:** The Address Reuse-Based Change Address Detection heuristic enhances the change address heuristic introduced by Meiklejohn et al.²⁰ by implementing an additional check on output address reuse of the labeled one-time change address.
11. **Paper:** Bitcoin address clustering method based on multiple heuristic conditions²²
 - (a) **Tool/Algorithm Outline:** He et al.²² present an improvement to the change address heuristic performance by adding additional constraints to the heuristic and improving the convergence speed of the algorithm. They present in detail some of the heuristics deployed on the Bitcoin network to cluster and track users.
 - (b) **Heuristic Description:** The improvement to the change address heuristic is well defined in a process diagram. The improvement includes additional checks to ensure that the address has not been used as an input before, there are only two output addresses in a transaction, and the amount of Bitcoin being transacted to the address under scrutiny has three more decimal places in comparison to the other address. When the transaction has more than two outputs, the output must meet the qualifications presented by Meiklejohn et al.²⁰

3.4 | Cluster intersections

The cluster intersection method takes identified inputs before they are obfuscated through a technique like a coinjoin and attempts to match those outputs later by linking them to the initially identified inputs.³⁵

In the Cluster Set intersection method, entity A controls two addresses A_1 and A_2 . Using heuristics like multi-input or change address, the cluster membership of both A_1 and A_2 is determined and belongs to the same cluster C_1 . Now, A uses some mixing techniques such as coinjoin to generate new transactions ct_1 with A_1 address as input resulting in Am_1 and ct_2 with A_2 resulting in Am_2 addresses, which are not linkable to A_1 and A_2 . Now, if A uses Am_1 and Am_2 in some payments to the same merchant/exchange, then the merchant can learn that both Am_1 and Am_2 form the cluster membership C_2 . By applying the intersection $C_1 \cap C_2$, one can establish that both A_1 and A_2 and Am_1 and Am_2 belong to entity A . One of the most significant drawbacks of this method is assuming the A is using a single wallet before

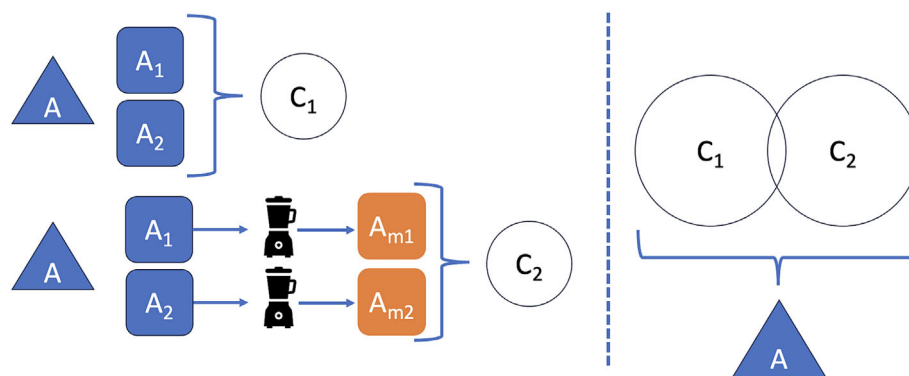


FIGURE 9 Figure showing the method as to how cluster intersection works by identifying outputs and assigning ownership based on co-spending of different cluster outputs simultaneously.

transaction ct_1 and ct_2 i.e., pre and post-cluster memberships are to be identified beforehand to make this scheme work. Figure 9 shows the process where four different outputs are generated by the same user. Two unmixed outputs are assigned to the first cluster, and the two mixed outputs are assigned to the second cluster. If the user, A , uses either of these outputs together where an intersection will occur, then the cluster intersection will define all of the outputs to be of the ownership of A .

Cluster intersection methods do not directly attack obfuscation methods like coinjoin in an attempt to identify the deterministic links between inputs and outputs, but rather attack the user's activity before and after a coinjoin.

3.5 | Fingerprinting

It is possible to extract information from the Bitcoin transactions from blockchain and non-blockchain sources. Transaction IPs can be extracted through studying of connectivity and traffic of peers,^{36,37} revealing possible locations of Bitcoin users. Even when users transact Bitcoin over a Tor network, they can still compromise their IP location.³⁸

Due to the increased use of smart phones for transacting Bitcoin,³⁹ it is possible to extract information through the wallet applications loaded on to user's phones. Aioli et al³⁹ demonstrated use of machine learning techniques applied to network traffic analysis on the applications housing Bitcoin wallets to identify user activity.

Biryukov and Tikhomirov⁴⁰ also demonstrate clustering techniques through the use of correlation on the timing of transactions on the network by analyzing mobile wallets. They also comment on the lack of privacy in Android wallets which could allow for the extraction of users' private information.

The time ranges of transactions can allow correlations to be made to identify possible mixing activity. Research provided by Wu et al¹⁹ show analyzing the possible time delay can narrow down the possibility of inputs and outputs to cluster in order to find the points of mixing in the network.

Wallet communication methods possess different patterns between a remote P2P node and a wallet depending on whether or not a previously received transaction belonged to the wallet. Biryukov and Tikhomirov⁴⁰ note that P2P wallets do not implement privacy broadcasting which can again be abused for the extraction of information on the users of the platforms.

Compiling the various fingerprinting information can provide enhanced insights into Bitcoin user activity, particularly in law enforcement, and can add as a potentially important feature in machine learning models.

3.6 | Machine learning driven detection

With publicly shared knowledge on the Bitcoin Wiki Privacy section⁴¹ on how to break the heuristics used by law enforcement or industry practitioners, the next wave of defense must come in the form of algorithmically driven detection methods through machine learning (ML) and deep learning (DL). Heuristics are rigid in their ability to

dynamically adjust the mechanics to keep up with the changing methods criminals or highly privacy-driven individuals use. ML and DL research are already capable of detecting malicious activity in social networks using digital currency,⁴² tax evasion identification,^{43,44} and e-commerce insurance fraud detection.⁴⁵ Using ML and DL techniques, it is possible to enhance the current heuristics deployed to deanonymize the Bitcoin network.

Research published by Gaihre et al⁴⁶ proposed the use of Graph Convolutional Networks (GCNs) in deanonymizing Bitcoin addresses. For a GCN to be deployed on Bitcoin, a graph must be established to capture the relations between Bitcoin addresses. As discussed in Section 3.7, how Bitcoin transactions are modeled is very important. For researchers to sufficiently identify addresses, we require the GCN to have a view of the individual inputs and outputs that compose Bitcoin transactions. However, a transaction can have up to 2000 inputs and outputs, which balloons the size of the dataset exponentially when compared with approx. 2700 transactions per block. Unpacking a block down to the input and output level could have approx. 2,000,000 rows. If GCNs are to identify relationships between addresses that can be dormant for weeks at a time, a highly scalable GCN is required.

Machine learning clustering techniques could aid in the probabilistic identification of the owners of multiple inputs in transactions. These methods can be coupled with the blockchain analytic graphical tools to prepare an investigative case. As a court requires a determinant outcome to link a suspicious person with specific outputs concretely, the ambition of a machine learning model is to complement current methods. Aiding a financial crimes analyst to sufficiently link the original target Bitcoin, for example, stolen Bitcoin, with an exit point like an exchange can be of great value for any justice institution. Some probabilistic models have been researched to compare against existing heuristics through historical testing.²³

3.7 | Transaction topology

The transaction topology of the Bitcoin network is the graph or network of interactions between the addresses that occur in transactions. Graphs play an important role in outlier detection and particularly within financial cybercrime¹ and allow us to capture relationships between data points.⁴⁷ Graphs are represented in a simple equation $G = (V, E)$, where G represents the graph dataset, V is a finite non-empty set also called the vertex or nodes set, and E are the pairs of distinct vertices or edges— E may be empty.

Typically, the Bitcoin network is modeled as the flow of cryptocurrency between users, where the edge set, E , is determined by the flow of Bitcoin from input A to output B . This area of research models off of the Elliptic Bitcoin dataset,⁴⁸ which has seen an massive increase in Graph Neural Networks being trained on a labeled dataset.^{49–53} The dataset is aggregated at a transactional level where the transaction is labeled illicit based on the underlying attribution of the inputs and outputs. High accuracy and evaluation metrics have been published on this dataset, but transforming the research for industrial application requires the combination of verifiable methods and reproducibility of results on out-of-sample datasets.

New research that has been developed by the same authors of this SoK restructures the transaction topology to focus on remodeling the edge set, E , through analysis of the similarity and affinity between illicit transactions on the network. Graph neural networks can be applied to the transaction graphs to discover the underlying representations that can allow computational models to discover other illicit transactions and malicious users. This new formulation of the Bitcoin edge set is shown to be an improvement with baseline GNNs when compared with the normal transaction based edge set.

A challenge in the field of graph analysis, particularly in cryptocurrency is the spatio-temporal factor which sees the network such as the Bitcoin transaction ledger expanding every ten minutes when a new block of approx. 2700 transactions is added to the blockchain.

3.8 | Anomaly detection

The goal of anomaly detection (AD) is to detect points of abnormality in a dataset by using various computational and mathematical techniques.¹ For many fraud problems, binary anomaly detection is the method of choice for deducing whether a transaction, account, or activity is deemed to be licit or illicit. Transaction data is classically treated as binary. It is inherently high dimensional and often contains highly correlated patterns.⁴⁷ Transaction data is typically temporal, with each transaction being assigned a time-stamp.⁴⁷

Some classical anomaly detection techniques include the following:

- *Density-based methods*: analyze density or sparsity of an area of data through parametric (i.e., Gaussian Distribution) or non-parametric techniques (i.e., kernel density). Kernel density estimation (KDE) creates a smooth density profile of the data population. The anomalies flagged are those that deviate from the estimated profile. It is presented mathematically as⁵⁴

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n \kappa\left(\frac{x - x_i}{h}\right) \quad (1)$$

where $\hat{f}(x)$ is the kernel density estimation at point x . n is the number of data points in the dataset. h is the bandwidth which the KDE smoothing parameter. Finally, $\kappa(\cdot)$ is the kernel function. The anomalies are detected by scoring data points x with $\frac{1}{\hat{f}(x)}$.

- *Network-based methods*: play an important role in anomaly detection within financial cybercrime. The state-of-the-art algorithms being published and researched on include either a preprocessing stage or direct analysis of the graph or network structures for identification of micro-clusters or sub-communities. An example of network or graph-based anomaly detection would be spatial proximity that models the use of links between nodes. One example would be betweenness centrality.⁵⁵ Betweenness centrality measures the extent to which a node lies on the shortest paths between other nodes in a network. It is represented mathematically as

$$C_B(v) = \sum_{s \notin v \notin t \in V} \frac{\sigma_{st}(v)}{\sigma_{st}} \quad (2)$$

where σ_{st} represents the number of shortest paths and $\sigma_{st}(v)$ is the number of shortest paths from s to t containing the vertex v . The anomalies can be detected through a threshold parameter, or continuous monitoring of a node's betweenness measurement can indicate a change in the graph dynamic and therefore emergence of anomalies.

3.8.1 | Defining normal Bitcoin transactions

Outliers in the context of Bitcoin have not been fully defined. As an outlier is also considered an abnormality, a normal transaction needs to be contextualized for any anomaly detection to appropriately occur. Some initial “normal” transactions can be removed from the dataset to decrease the scope of transactions being monitored. Coinbase transactions, which pay the miner of the block the Bitcoin reward plus transaction fees, can be considered a normal transaction in the context of Bitcoin. Bitcoin mining rewards are likely to be paid out to mining pools. Identifying the subsequent payment out to the network will identify all associated miners' addresses in the Bitcoin network dedicating their processing power. Other transactions that should be of interest is defining investment transactions. These transactions would occur between an exchange, like Coinbase or Binance, where a user wants to buy Bitcoin and hold it for investment purposes. These users do not participate in the exchange of Bitcoin between other Bitcoin users but interact exclusively with the exchange and possibly another wallet to protect their Bitcoin funds. Identifying these small transactions between single wallets and exchanges in two way interactions could eliminate a large pool of investment participants in the Bitcoin network. There is difficulty in this method, however, as exchanges do not typically send out individual transactions as they would incur high transaction fees. Instead, they batch multiple orders together and send out many inputs to multiple customer output addresses. Identifying these batch transactions could aid in identifying investment transactions if the receiving address is static for a certain amount of time.

This leaves the true normal transactions between users in the network. Anomalous transactions in terms of exchanges or law enforcement could be the identification of users interacting or engaging with obfuscation techniques including mixing (Section 2.3.1), coinjoins (Section 2.3.2), or peeling (Section 2.3.3). To identify these abnormal transactions in a network, it is necessary to create a class labeling structure. Labeling Bitcoin transactions as illicit or licit based on the underlying properties of the input and output addresses and whether the transaction seems to be engaging in obfuscation can allow us to perform supervised learning tasks. The difficulty in aggregating transaction data lies within the formulation of the data structure. Unpacking a Bitcoin block unloads approximately 2700 transactions, where each input and output is represented on a row. To create an aggregated transaction table one must roll back up the inputs

and outputs into a structure that represents approx. 2700 transactions in a block without losing the underlying information on the inputs and outputs.

3.9 | Evaluation metrics

Across cryptocurrency anomaly detection research, the F1-score evaluation metric is reported regularly as it captures a model's performance when being applied to highly imbalanced class datasets with a binary class target.

Accuracy is a method of evaluating the performance of a binary classification model distinguishing between True or False. Accuracy is determined by summing the correctly classified outputs (True Positive and True Negative) and dividing by the total number of classifications which is True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). Equation (3) below represents the accuracy calculation:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

Recall is considered a percentage of ground-truth anomalies, which have been classified as anomalies at a given threshold.⁴⁷ Equation (4) calculates the recall metric.

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

Precision is considered to be the percentage of reported anomalies, which are actually true anomalies.⁴⁷ Equation (5) calculates the precision metric.

$$Precision = \frac{TP}{TP + FP} \quad (5)$$

The F1-score is the harmonic mean of both precision and recall. Equation (6) calculates the f1-score.

$$F1\text{-score} = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (6)$$

3.10 | Law enforcement requirements

Computational models deployed with the ambition of assisting law enforcement in the identification of criminal individuals or entities must provide particular requirements to law enforcement agencies in order for their findings to be admissible in court. Van Baar et al⁵⁶ discuss the digital forensics process with regards to the aiding of law enforcement, including that any evidence supplied must be done so through verified methods.

Law enforcement agencies require determinant output, which is why heuristics are heavily leaned upon for the aid of financial crime analysts. Probabilistic output can be argued against in a court of law. However, if the ambition of a model is to purely aid an investigator by supplying the likelihood of a transaction, output, or input of being illicit, then a probabilistic model can be used. With regard to heuristic used in cryptocurrency to deanonymize and cluster actions of suspects, Frowis et al⁵⁷ discuss the possibility of super clusters being created by heuristics which can impact the forensic evidence provided by investigators.

4 | PRIVACY ENHANCING TOOLS

Many tools being made available to criminals enhance their ability to further obfuscate their activity online. There is a drive from privacy-focused individuals and malicious actors to provide avenues of heightened privacy and identity

protection, however, these tools produced by these specialists allow for the hindrance of justice and directly aid in the ability of criminals to launder their money more efficiently. Privacy-focused individuals criticize the use of Know Your Customer (KYC) by exchanges, who have been accused of erroneously blacklisting wallets and seizing cryptocurrency until a dispute is settled. This potential misclassification of licit activity results in punishment that has repercussions on the network. This is covered in the work by Kolachala et al.,⁸ who discuss the challenges in anti-money laundering in blacklisting coins on a cryptocurrency network. The adverse response by malicious actors and genuine privacy advocates has spawned advanced computational tools to help obfuscate transactions. In this section, we detail tools and information available to Bitcoin users which allow them to significantly enhance their anonymity and privacy while using Bitcoin for illicit activity.

4.1 | Antinalysis

Blockchain analytical software called Antinalysis⁵⁸ has become freely available for Bitcoin and Monero users to risk-assess their wallets with the ambition of evading law enforcement and exchanges identifying their historical activity.

Antinalysis gives a history up to 5-hops from the input address. There is a generalized consensus in Bitcoin privacy advocates that exchanges only have the capability of examining 5-hops for the history of a wallet, so users are encouraged to obfuscate their history beyond 5 transactions.

Tools like Antinalysis give criminals an edge when engaging with exchanges. This will allow them to evade any blacklisting, assess the performance of their mixing, coinjoin, or peeling methods. To benefit users of the tool, the Antinalysis team only retain look-up searches for 72 h and then delete their database.

4.2 | Privacy-driven wallet providers

Privacy enhanced wallets have become available to all users of cryptocurrency networks, many of them freely available. These products advertise their ability to mask users' identity on the network. A study by Ghesmati et al. examined the usability of wallets providing coinjoin services.⁵⁹ They studied Samourai, Wasabi, and JoinMarket. The main criticisms were around the ease of use for the applications, where inexperienced users would have difficulty executing the coinjoin operations. Also, the use of previously mixed inputs and other UTxO which is belonging to the same user can still be clustered through the multi-input heuristic.

Whirlpool, a product offered by Samourai, examine the deterministic link within transactions and their originating inputs. Boltzmann entropy is the core of their entropy measurement, with the goal of avoiding over clustering the unspent transaction output when performing coinjoin operations. Their GitHub is accessible, and full discussion on improving techniques is publicly available.⁷

4.3 | Exchanging crypto and FIAT

Another avenue of entering cryptocurrency networks with the transfer of FIAT money other than standard exchanges is accessing Bitcoin ATM networks.⁶⁰ These networks have varying conditions regarding the documentation a person must supply in order to lodge cash. Using www.coinradaratm.com allows one to search all known cryptocurrency ATMs worldwide. Included in this website is information surrounding each ATM where available. Some ATM details surrounding the requirement of verification documents on limits up to \$1000, where deposits can be made to wallets without any KYC requirement, are also available. This could allow criminals to engage in “smurfing” activities across a wide range of ATMs to launder criminal proceeds. The ATMs provide a method of obtaining cryptocurrency into a completely anonymous account which can be used for the purchase of illicit goods or services, or simply launder the deposit of criminal proceeds and sell it back to exchange under the guise of a cryptocurrency trade.

Cryptocurrency video games are another avenue of laundering criminal proceeds with the “play-to-earn” model. A criminal can enter their criminally obtained cryptocurrency into the games and transfer their “earned” cryptocurrency to new wallets and or directly exit via an ATM⁶² under the guise of an earned reward.

5 | RECOMMENDATIONS

In this section, we present our recommendations to the research community. For research to improve, labeled data released by leading entities could push the next generation of illicit identification. Similarly, enhancing the discussed heuristics with artificial intelligence through ML and DL techniques can drastically improve the current identification methods. Finally, we discuss the expert cyber criminal.

5.1 | Public data

Bitcoin itself is publicly accessible, but without any labels on the wallets of the controlling entity or datasets of digestible illicit activity on the network, there are huge limitations to identifying bad actors.

The current datasets available to aid researchers in identifying illicit activity are extremely limited.

The Elliptic Bitcoin graph dataset^{48,63} is one that the community has thoroughly researched. The graph dataset is temporal, and the data can be split over 49-time steps. The nodes have been either labeled licit (42,019 nodes or 21%), illicit (4545 or 2%), or unknown (157,205 or 77%). The labels were informed using a heuristics-based reasoning process.⁴⁸ A massive drawback to the dataset is its anonymity and assumption that the heuristics have appropriately labeled the nodes as licit or illicit. There is also a limitation to the applicable research as only a small sample of nodes is labeled. The removal of the unknown nodes, performed by researchers who apply deep learning models to the dataset, most likely has a direct model performance impact as relational data points are lost, and graphs are changed.

The end goal of any money laundering scheme is to integrate criminal funds into the public financial system and have funds deemed legitimate by law enforcement. Criminals will purchase precious metals and luxury goods, invest in property and vehicles, or extract FIAT currency from a legitimate exchange on the internet and class any profit as trading activity. One of the most common exit points in any of the aforementioned cryptocurrency transaction methods is an exchange like Coinbase or Binance. In many high-profile cases, criminals are identified through the KYC process customers must go through to verify their identity to comply with regulations.

There are improvements in the authentication steps required by the exchanges for new customers, including government identification, and small video recording required to ensure that the user is not using any fake documents. However, there is still the ability of sophisticated criminals to overcome any defense that does not require in-person verification.

Our recommendation for the enhanced protection of the cryptocurrency network is for exchanges to publish lists of wallets labeled as exchange points for researchers and law enforcement to analyze the blockchain appropriately and analyze multiple hops previous to the exchange point. This will allow researchers to test new methods and evaluate current state-of-the-art techniques being deployed extensively. Another avenue of research is combining machine learning techniques with fingerprinting, which can capture users' activity. Combining fingerprinting data with network topology could be a powerful source of relationship identification through sub-community detection to potentially discover groups of malicious actors.

5.2 | Artificially intelligent heuristics

The assumptions underlying the heuristics that attempt to deanonymize, cluster, and identify addresses are evaded by the sophisticated techniques deployed by advanced users of Bitcoin. Multi-input address heuristic trying to cluster inputs by assuming all inputs are sent by the same user are instantly broken through coinjoins, as shown in Figure 5. The only scenario where this can be genuinely identified as a multi-input by the same user is if the address hash of each input is the same, or verifiable attribution on the addresses has been collected. The multi-input heuristic will simply not label addresses that are not co-spent with another address.⁶⁴ Similarly, change addresses can only be identified if the wallet address is the same or an assumption is made that the one of the two outputs belongs to the sender. For example, when Alice sends one input to two outputs as seen in the second scenario of Figure 1. Additionally, batch transactions performed by mining pools and exchanges to reduce transaction fees directly affect the underlying assumptions with change address detection.

Heuristics provided researchers with rich datasets early in the use of Bitcoin. Still, as the user base and supporting developers have grown more sophisticated in their use of Bitcoin, the heuristics need to be dramatically improved.

Heuristics have a very high error rate, as published by Gong et al.⁶⁵ The multi-input yields an average error rate of 63.46% and the change address heuristic 92.66%. Gong et al. also recommend the use of more advanced technology, which can allow for the replacement of poor-performing heuristics.

Batch transactions are an intuitively more intelligent method to send transactions in the network by minimizing transaction fees to miners, which goes against any change in address identification. There is still merit in using heuristics to identify the more basic users of Bitcoin who are not trying to mask their activity. Still, other methods must be deployed on top of the existing heuristics to ensure their identification.

There is strides being made in an attempt to apply ML to deanonymize Bitcoin users. Research has been deployed in applying unsupervised methods to cluster miners in the Bitcoin network through analysis of similar mining hash rates.⁶⁶ Early machine learning methodology applied to Bitcoin deanonymization included work by Ranshous et al.³⁴ In their study, their motivation is to aid in anti-money laundering by identifying patterns unique to exchange controlled addresses. They demonstrated the effectiveness of machine learning models in Bitcoin such as random forest and AdaBoost to show extremely high evaluation metrics in identifying exchange addresses.

As mentioned in Section 3.3, work by Jourdan et al.³³ used logistic regression and Light Gradient Boosting Machine (LGBM) models to classify various entities that controlled Bitcoin addresses; this was coupled with heuristic modeling.

Work performed by Möser and Narayanan⁶⁴ attempting to apply a random forest classifier, among others tested, to predict a transaction's change output. There is still a lot of opportunity in the graph neural network space to aid in learning embeddings that represent change address transactions or multi-inputs and have their output complement existing heuristics. An extension of the work by Möser and Narayanan⁶⁴ was performed by Kappos et al.⁶⁷ in their analysis of random forest application to the identification of peel chains.

Our recommendation to the research community is to combine probabilistic modeling with the existing heuristics to appropriately determine the ownership of the inputs and outputs for transactions. As shown in the cluster intersection method, a direct attack of the obfuscation methods is not required, but determining users before and after these methods can gain massive insights. More specifically, analyzing new techniques in the transaction topology space can yield vital insights into sub-community identification between malicious actors and reveal wider groups of cybercriminals.

5.3 | Tackling the expert criminal

A dichotomy exists between privacy and safety on the Bitcoin network. As academics continually publish research pushing the field of cryptocurrency to enhance users' privacy, the results have lasting effects on the cryptocurrency system. The more private a network is made, the more likely it is to be labeled a haven for illicit behavior, and indeed does attract cybercriminals and organized crime gangs to launder their funds more increasingly over time.⁶⁸

Similarly, the tooling ecosystem has matured both commercially and open source software to help law enforcement track and investigate criminals leveraging Bitcoin networks for illicit gains. When collected as evidence, cryptocurrency addresses associated with a crime can play a vital role both in pre-trial and during the trial of criminal proceedings. Tools based on heuristics and clustering methods, when used to link Bitcoin addresses to a person under investigation and presented as evidence, need to document and report the methods leveraged to establish the links and disclose the reliability metrics and labeling techniques. Black box treatment of these tools may result in uncertainty in establishing the required degree of suspicion and hinder admitting as scientific evidence in criminal proceedings.

Expert criminals take advantage of the weaknesses of the heuristics and black box models to provide a level of uncertainty and plausible deniability between them and outcomes of models. To be confident in the produced model output by the research community, reproducibility and verifiable results must be of focus which guarantee the authenticity of results.⁵⁷

6 | CONCLUSION

In this systemization of knowledge, we have provided readers with a comprehensive and digestible list of obfuscation methods, illicit activity, and deanonymization mining methods including an extensive review of the heuristics deployed on the network, covered several privacy enhancing methods used by Bitcoin users, and provided recommendations surrounding the addressing of identifying advanced Bitcoin users.

As cryptocurrency increases in its use by cybercriminals for the laundering of funds, so too must the research for combating the illicit activity. Like any cybersecurity problem, the advantage of attack and defender changes over time. Cybercriminals have an advantage of using tools to enhance their privacy and monitor their own activity to evade detection by exchanges when transferring cryptocurrency to FIAT currencies.

Artificial intelligence techniques through machine learning and deep learning have shown huge promise in detecting money laundering in FIAT currencies. There is some adoption to the cryptocurrency world but as data is far more limited than what is available for FIAT, it is difficult to evaluate the performance of these models. However, combining ML and DL techniques with the current heuristics being applied to the Bitcoin network could show drastic improvements.

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